Deep Learning-Based Multi-User Detection in Underwater Optical NOMA Networks

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Abstract—Underwater Optical Wireless Communication (UWOC) has emerged as a promising technology for high-speed underwater data transmission required for ocean monitoring, autonomous underwater vehicles (AUVs), offshore exploration, and environmental sensing. However, optical communication in underwater environments suffers from severe attenuation, scattering, and turbulence caused by oceanic particles and temperature gradients. Non-Orthogonal Multiple Access (NOMA) allows multiple underwater users to share the same optical spectrum through power-domain multiplexing. Traditional receivers rely on Successive Interference Cancellation (SIC) for decoding, but their performance deteriorates significantly in highly turbid underwater channels. This paper investigates a deep learning-based receiver for multi-user detection in UWOC-NOMA systems. A 1-Dimensional Convolutional Neural Network (1D-CNN) is implemented to decode superimposed signals transmitted through simulated underwater channels modeled using the Beer–Lambert law and lognormal turbulence. MATLAB simulations compare the Bit Error Rate (BER) performance of the CNN-based decoder with the conventional SIC receiver. Results demonstrate that the proposed CNN-based receiver maintains lower BER values over longer communication distances and under high turbidity conditions, confirming the effectiveness of deep learning techniques in improving the reliability of underwater optical communication systems.

Index Terms—*Underwater Optical Communication, NOMA, Deep Learning, CNN, Beer-Lambert Law, Multi-User Detection, MATLAB Simulation.*

I. INTRODUCTION

Underwater communication plays a crucial role in numerous applications including oceanographic data collection, environmental monitoring, underwater robotics, military surveillance, and offshore exploration. Traditional underwater communication systems primarily rely on acoustic waves due to their long propagation range in water. However, acoustic communication suffers from several limitations such as low data rates, high propagation delay, limited bandwidth, and vulnerability to environmental noise.

Underwater Optical Wireless Communication (UWOC) has emerged as an attractive alternative capable of supporting high-speed data transmission with low latency. Optical communication systems use visible or near-infrared light sources such as LEDs or lasers to transmit data underwater. Compared to acoustic systems, optical communication can provide significantly higher data rates, making it suitable for applications involving real-time video transmission and high-bandwidth sensing.

Despite its advantages, UWOC systems face major challenges due to the complex nature of the underwater channel. Light propagation underwater is affected by absorption, scattering, and turbulence caused by suspended particles, salinity variations, and temperature gradients. These phenomena significantly degrade signal quality and limit communication range.

To improve spectral efficiency and support multiple underwater devices simultaneously, Non-Orthogonal Multiple Access (NOMA) has been proposed for optical communication systems. NOMA allows multiple users to share the same communication channel by allocating different transmission power levels. At the receiver, Successive Interference Cancellation (SIC) is used to separate the superimposed signals.

However, SIC relies on accurate channel estimation and simple mathematical decision rules, which may not perform well in highly turbulent underwater environments. Recently, deep learning techniques have been successfully applied in wireless communication systems to learn complex channel characteristics and perform signal detection without requiring explicit mathematical channel models.

This paper explores the use of a deep learning-based receiver to decode NOMA signals transmitted through underwater optical channels. A convolutional neural network (CNN) is trained to directly recover transmitted bits from received optical waveforms, improving robustness against channel distortions. The remainder of the paper is organized as follows: Section II reviews related literature; Section III describes the system architecture; Sections IV and V detail the transmission and channel models; Section VI presents the CNN-based receiver; Sections VII–IX cover dataset generation, simulation parameters, and results; Section X concludes the paper.

II. LITERATURE REVIEW

Recent research has shown growing interest in underwater optical wireless communication due to its ability to provide high-speed connectivity for underwater networks. Zeng et al. [1] presented a comprehensive survey of UWOC systems highlighting the importance of modeling optical attenuation using the Beer–Lambert law and turbulence using statistical models such as lognormal or Gamma-Gamma distributions.

NOMA has been widely studied in radio frequency communication systems as a technique to improve spectral efficiency. Dai et al. [2] demonstrated that NOMA allows multiple users to access the same frequency resources simultaneously using power domain multiplexing. In optical wireless communication systems, researchers have investigated the application of NOMA to improve network capacity. However, the performance of NOMA systems heavily depends on the effectiveness of the receiver algorithms used to separate superimposed signals.

Traditional receivers rely on Successive Interference Cancellation (SIC), which decodes the strongest user first and then subtracts its contribution from the received signal before detecting

the next user. While SIC works well in ideal conditions, its performance degrades in channels with strong noise, fading, or nonlinear distortions.

Deep learning techniques have recently been introduced into the physical layer of communication systems. O'Shea and Hoydis [3] demonstrated that neural networks can learn optimal signal detection strategies directly from data. Convolutional neural networks and autoencoders have been applied to modulation recognition, channel estimation, and signal detection tasks. Combining UWOC channel modeling with deep learning-based multi-user detection remains an emerging research area, which this paper aims to address.

III. SYSTEM ARCHITECTURE

The proposed communication system consists of the following major components: (1) random bit generation for multiple users; (2) BPSK modulation; (3) NOMA superposition coding; (4) underwater optical channel modeling; (5) signal reception and decoding; and (6) performance evaluation using BER metrics.

The transmitted signal from two users is combined using power domain NOMA and then propagated through the simulated underwater channel. At the receiver, both a traditional SIC decoder and a CNN-based decoder are implemented and compared. The overall system workflow follows the pipeline:

Transmitter → NOMA Superposition → Underwater Channel → Receiver → Decoder → BER

Fig. 1 illustrates the block diagram of the proposed system. The figure is omitted here pending actual MATLAB simulation output but should include: Transmitter block, BPSK Modulator, NOMA Superposition Encoder, Underwater Optical Channel block (Beer–Lambert + Turbulence + AWGN), Receiver block, CNN Decoder / SIC Decoder, and BER Calculator.

[Fig. 1. Proposed Underwater Optical NOMA Communication System Architecture]

IV. NOMA TRANSMISSION MODEL

In power-domain NOMA systems, signals from multiple users are transmitted simultaneously with different power levels. For two users, the transmitted superimposed signal is given by:

$$X_tx = \sqrt{P_1} \cdot x_1 + \sqrt{P_2} \cdot x_2 \qquad (1)$$

where x_1 is the BPSK-modulated signal of User 1 (far user), x_2 is the BPSK-modulated signal of User 2 (near user), P_1 is the power allocated to the far user, and P_2 is the power allocated to the near user. Since the far user experiences larger attenuation, it is assigned higher transmission power. The power allocation used in this work is $P_1 = 0.8$ and $P_2 = 0.2$, ensuring $P_1 + P_2 = 1$.

V. UNDERWATER OPTICAL CHANNEL MODEL

A. Beer–Lambert Attenuation

The attenuation of optical signals propagating through water is modeled using the Beer–Lambert law [1], [7]. The channel attenuation coefficient h_l is given by:

$$h_l = \exp(-c \cdot d) \quad (2)$$

where c is the attenuation coefficient of the water medium (m^{-1}) and d is the transmission distance (m). The attenuation coefficient depends on the water type and particle concentration. Based on the Jerlov water classification [7], representative values are provided in Table I.

TABLE I

Attenuation Coefficients for Different Water Types

Water Type	Attenuation Coefficient c (m^{-1})
Pure Sea Water	0.056
Clear Ocean Water	0.15
Coastal Water	0.398

B. Optical Turbulence Model

Underwater turbulence caused by temperature gradients, salinity fluctuations, and air bubbles introduces random fluctuations in received signal intensity. This effect is modeled using a lognormal random variable T, a common statistical model for weak-to-moderate turbulence [1]. The received signal is expressed as:

$$Y = h_l \cdot T \cdot X_{tx} + n \quad (3)$$

where T is the lognormal turbulence factor and n is the additive white Gaussian noise (AWGN) with zero mean.

VI. DEEP LEARNING-BASED RECEIVER

A. Motivation

Traditional SIC receivers perform sequential decoding based on threshold detection. However, underwater optical channels introduce complex and nonlinear distortions that degrade SIC performance, particularly at longer distances and under high turbidity. Deep learning models can learn complex channel behaviors and directly map received signals to transmitted bits without requiring explicit channel estimation or mathematical inversion.

B. 1D-CNN Architecture

A 1-Dimensional Convolutional Neural Network (1D-CNN) is employed as the AI-based multi-user detector. The network architecture is as follows:

1) Input Layer — Received signal samples
2) 1D Convolutional Layer — 16 filters, ReLU activation
3) 1D Convolutional Layer — 32 filters, ReLU activation
4) Fully Connected Layer
5) Sigmoid Output Layer — binary outputs for each user

The network outputs two binary sequences corresponding to the decoded bits of User 1 and User 2. The use of convolutional layers allows the network to exploit local temporal correlations in the received waveform, improving detection robustness.

VII. DATASET GENERATION AND TRAINING

The training dataset is generated entirely through MATLAB simulation following these steps: (1) generate random binary data for two users; (2) apply BPSK modulation; (3) apply NOMA power-domain superposition; (4) pass the signal through the simulated underwater optical channel; (5) store the received signal as the network input; and (6) store the original transmitted bits as labels.

The dataset is partitioned as: Training Set (70%), Validation Set (15%), and Test Set (15%). This split ensures proper generalization of the neural network and mitigates overfitting. The Adam optimizer [5] with a batch size of 128 is used for training over 20 epochs.

VIII. SIMULATION PARAMETERS

TABLE II

Simulation Parameters

Parameter	Value
Number of Users	2
Modulation Scheme	BPSK
Power Allocation	P1 = 0.8, P2 = 0.2
Distance Range	1 - 30 m
Turbidity Levels	3 (Pure Sea, Clear Ocean, Coastal)
Channel Model	Beer-Lambert + Lognormal + AWGN
Neural Network	1D-CNN

Training Epochs	20
Batch Size	128
Optimizer	Adam

IX. RESULTS AND DISCUSSION

A. Eye Diagram Analysis

Eye diagrams are used to evaluate signal quality at the receiver. For pure sea water, the eye diagram remains widely open, indicating low distortion and good signal integrity. In coastal water environments, strong scattering and attenuation cause significant eye closure. This makes threshold-based detection methods such as SIC less effective and increases the probability of bit errors. Fig. 2 and Fig. 3 illustrate the eye diagrams for pure sea water and coastal water conditions, respectively.

[Fig. 2. Eye Diagram — Pure Sea Water] *[Fig. 3. Eye Diagram — Coastal Water]*

B. BER vs. Distance Performance

Simulation results demonstrate that BER increases monotonically with transmission distance due to increased Beer–Lambert attenuation. The traditional SIC receiver begins to fail beyond approximately 15 meters in highly turbid (coastal) water, as its threshold-based decision rule is unable to compensate for the complex nonlinear distortions introduced by the channel.

The CNN-based receiver maintains significantly lower BER values even at distances up to 25 meters, demonstrating its ability to implicitly learn and compensate for complex channel distortions. This performance advantage is most pronounced under coastal water conditions where turbidity is highest. Fig. 4 illustrates the BER vs. distance comparison for both receivers under the three water types.

[Fig. 4. BER vs. Link Distance for SIC and CNN Receivers (semilogy scale)]

C. CNN Training Convergence

The training curves show consistently decreasing loss and increasing validation accuracy over the 20 training epochs. This confirms that the 1D-CNN successfully learns the statistical characteristics of the underwater optical channel from the simulated training data. No significant overfitting was observed, validating the 70/15/15 dataset split strategy. Fig. 5 shows the training loss and validation accuracy curves.

[Fig. 5. CNN Training Convergence: Loss and Validation Accuracy vs. Epochs]

X. CONCLUSION

This paper presented a deep learning-based multi-user detection technique for underwater optical NOMA communication systems. The underwater channel was modeled using Beer-Lambert attenuation combined with lognormal turbulence and AWGN noise, covering three representative water types. A 1D-CNN receiver was implemented and trained on MATLAB-generated simulation data to decode superimposed NOMA signals.

Simulation results demonstrated that the CNN-based decoder significantly outperforms the conventional SIC receiver in terms of BER performance, particularly in highly turbid water and long-distance communication scenarios. These results highlight the potential of deep learning techniques to enhance the reliability and efficiency of next-generation underwater communication networks.

Future work will include: extending the system to support more than two simultaneous users; investigating advanced deep learning architectures such as LSTM or Transformer networks; conducting real-time UWOC hardware experiments; and exploring adaptive power allocation strategies that jointly optimize NOMA and the neural network decoder.

XI. REFERENCES

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